



POLITECNICO
DI TORINO

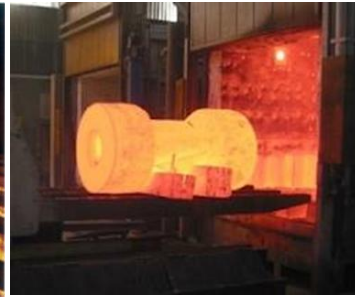
Technology of Metallic Materials A.Y 2025-26

Mechanical Engineering

Steels: Modern Production



Encarta Encyclopedia, Farrell Grehan/Photo Researchers, Inc.



Giovanni Maizza
DISAT

Lecture 8

1



POLITECNICO
DI TORINO

Production of Steels

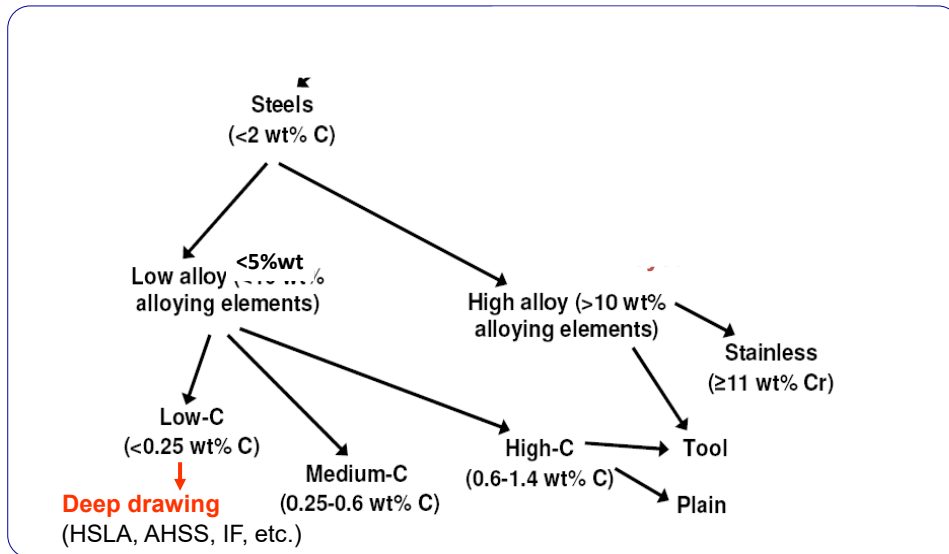
1. **Integral production** (minerals):
 - blust furnace production → pig iron → Cast iron
 - BOF-converter+ pig iron → Steels
2. **Secondary production** (scraps+ pig iron): BOF → Steel
3. **Direct production**: electric furnace (scraps) → Alloy Steels (Special steels)

2



POLITECNICO
DI TORINO

Classification of Steels

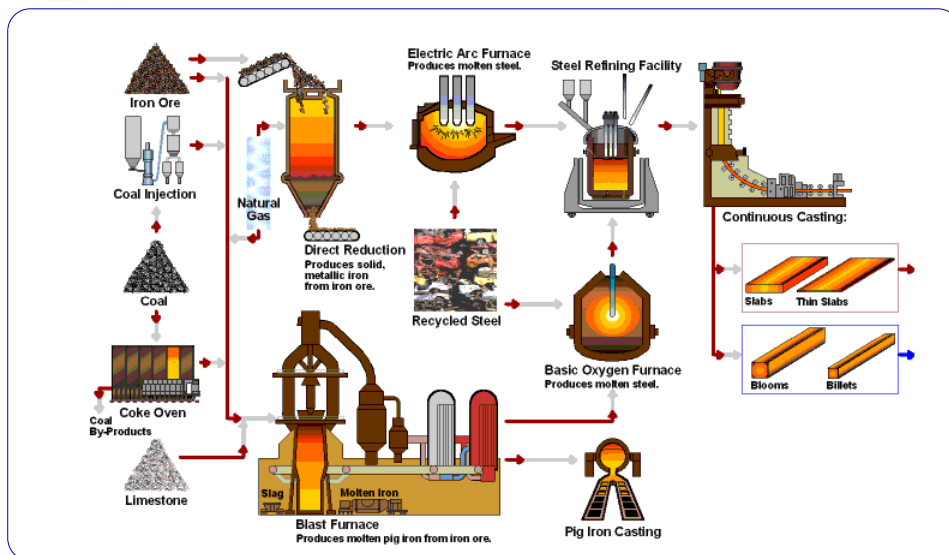


3



POLITECNICO
DI TORINO

Integral Production of Steels

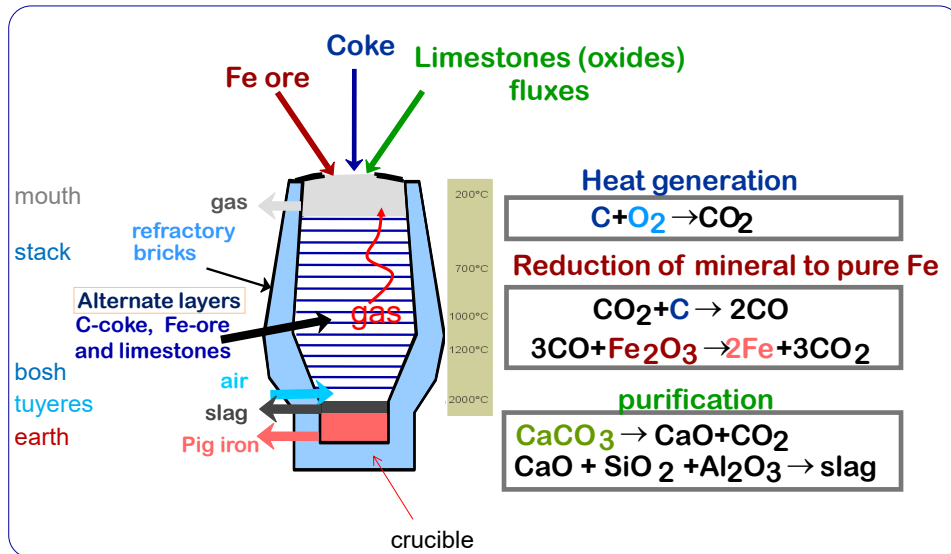


4



POLITECNICO
DI TORINO

REACTIONS IN THE BLAST FURNACE

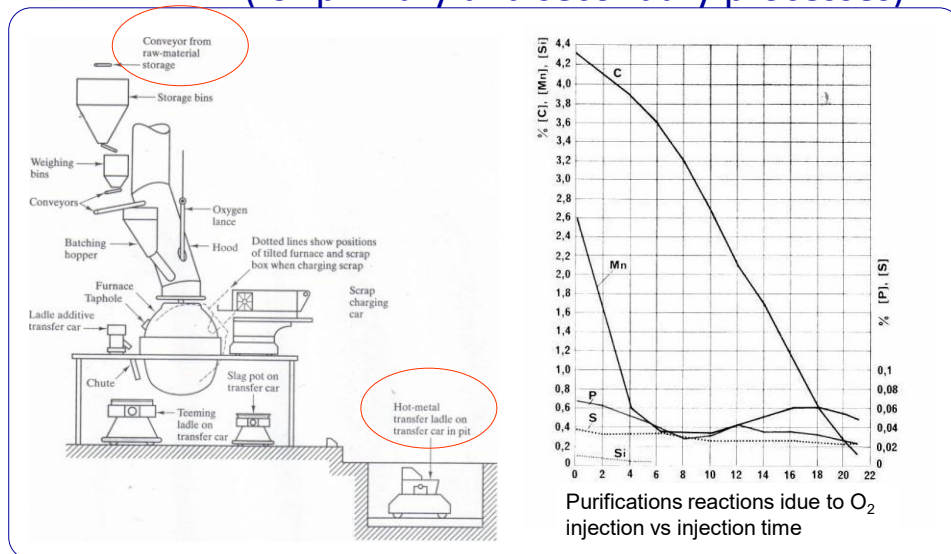


5



POLITECNICO
DI TORINO

BOF Operations (for primary and secondary processes)



6



POLITECNICO
DI TORINO

Harmful Impurities

Max tolerable content of impurities in high quality steels :

S= 0.03%; P= 0.02%, Cu=0.3%

Sn=0.06%; Ni=0.3%; Cr=0.2%

NB. S and P are detrimental for all steel grades except for **rephosphorized** and **resulfurized steels** in which a deliberate amount of these element is added.

Si and **Mn** are always present because contribute to melt purification.

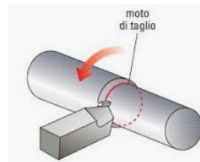
P : embrittlement due to segregation of Fe_3P ,

S : hot embrittlement due to segregation of FeS at GBs

Resulfurized steels (for automatic machining):

S < 0.06-0.35 %

S: promotes brittle chips for easier removal from the work



7



POLITECNICO
DI TORINO

Purification of Steels - Deoxidation

After decarburizing a subsequent step is necessary to **minimize entrapped O** in the melt (**Degasing process by Ar gas or the more costly high vacuum technology**). However, some O residual will remain.

Even 100 ppm of **O** or **N** are sufficient to induce **embrittlement**.

In more advanced technologies **deoxydation** may combine with a decarburizing operation by adding **deoxydizing elements as ingots of:**

- Cast iron (C)
- Fe-based alloy enriched of Si, Mn, (Cr, expensive)
- Al-Si alloys, Al ingot

as Al, Si, Mn (decreasing order) exhibit high chemical affinity to O (and N)

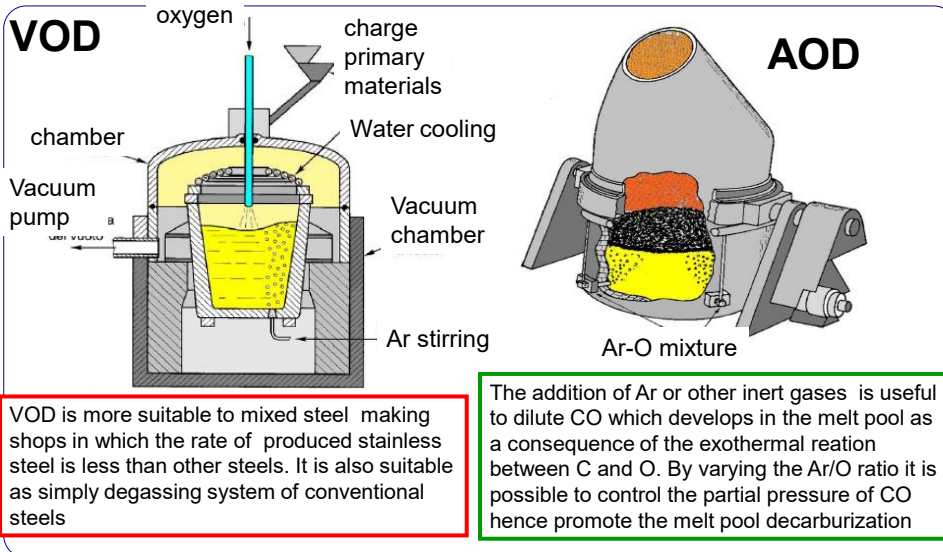
Note that: C in Fe-based alloys acts as a deoxydizing element. Al may block N and O by forming very hard AlN and Al_2O_3 precipitates respectively and Mn can block O and S in the form of inclusions

8



POLITECNICO
DI TORINO

PRODUCTION OF STAINLESS STEELS

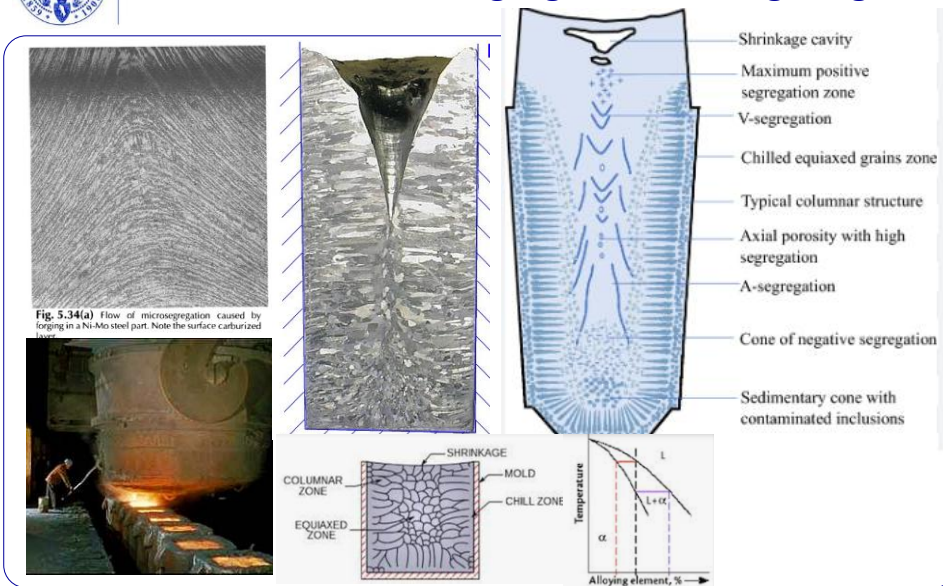


9



POLITECNICO
DI TORINO

Macrosegregations in large Ingots



04/12/2025

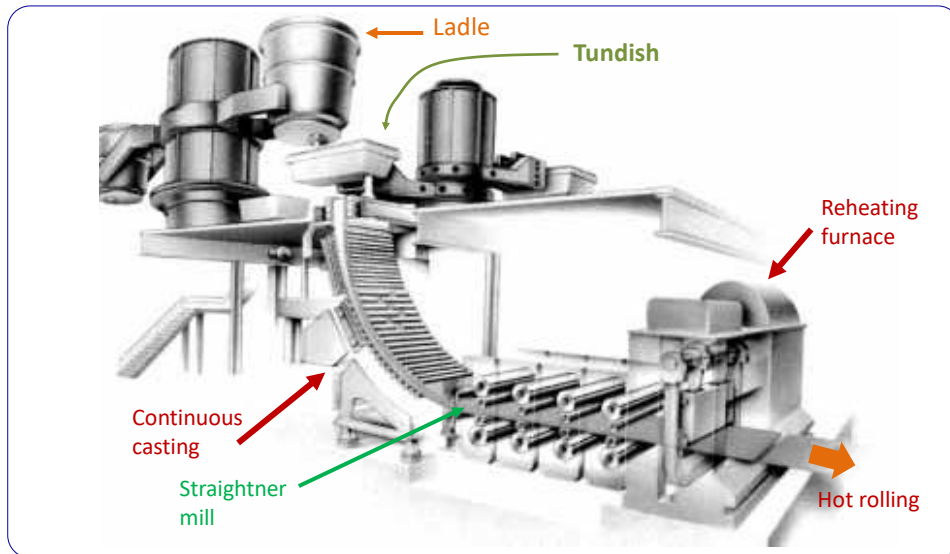
Mechanical Engineering

10



POLITECNICO
DI TORINO

3D view of continuous casting and hot rolling



11



POLITECNICO
DI TORINO

Hot rolling Work



An innovative hot rolling mill, featuring the "pair-crossed roll" system, at Hirohata Works

Nippon Steel: Hirohata Works

12

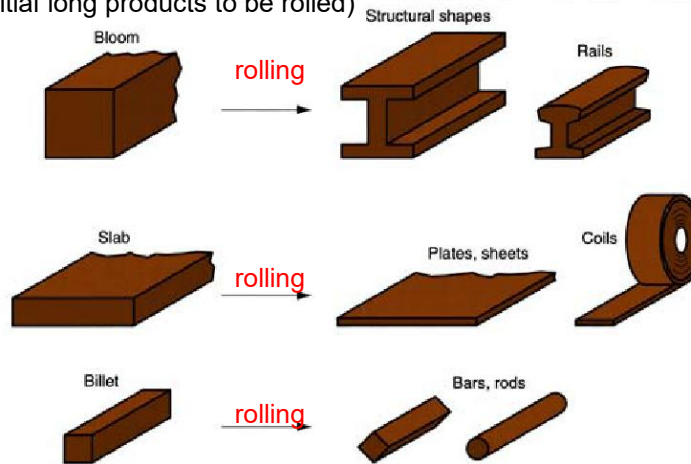


POLITECNICO
DI TORINO

Hot-rolling Intermediates

Metallurgical intermediates
(=initial long products to be rolled)

Finished hot rolled products



04/12/2025

Mechanical Engineering

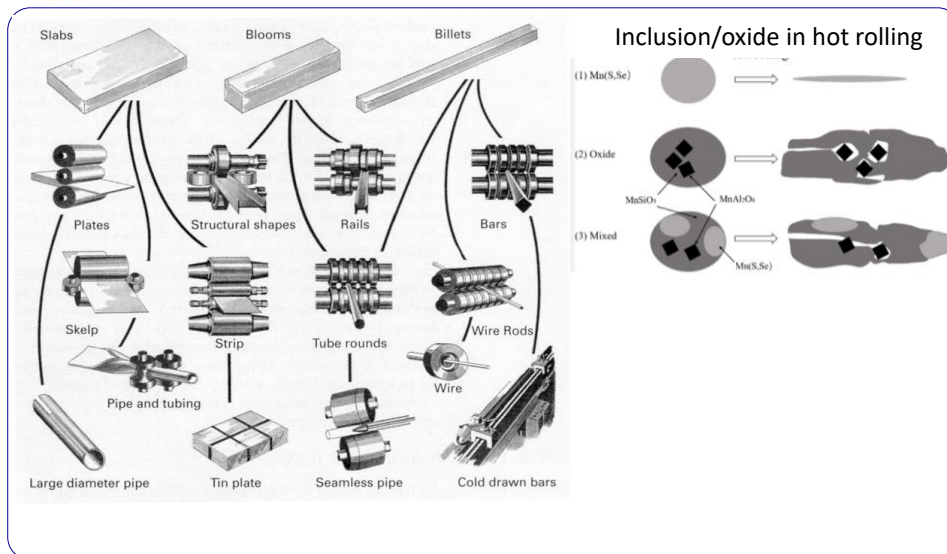
13

13



POLITECNICO
DI TORINO

FINISHED AND SEMI-FINISHED HOT ROLLED PRODUCTS

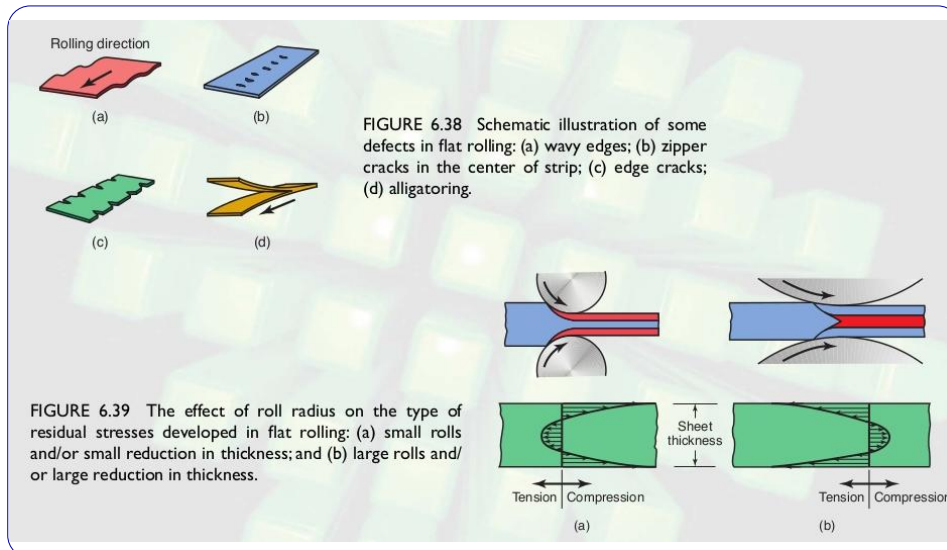


14



POLITECNICO
DI TORINO

Rolling defects and Surface stresses



04/12/2025

Mechanical Engineering

15

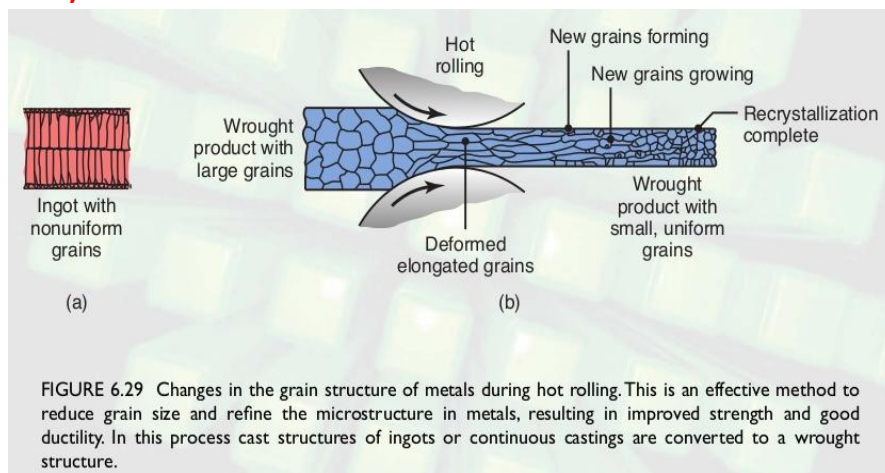
15



POLITECNICO
DI TORINO

Metallurgical Phenomena during hot rolling

Recrystallization



04/12/2025

Mechanical Engineering

16

16

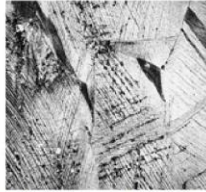


POLITECNICO
DI TORINO

Recrystallization phenomena vs. Time/Temperature



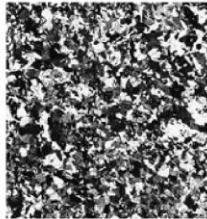
Cold-worked (33%CW)



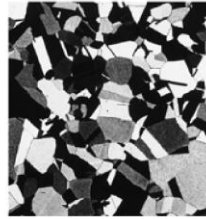
Initial stage of rec. after
heating 3 s at 580°C



Partial replacement of cw
grains by rec. ones 4 s.



Complete recrystallization
(8 s at 580°C).



Grain growth after 15
min at 580°C



Grain growth after 10
min at 700°C

04/12/2025

Mechanical Engineering

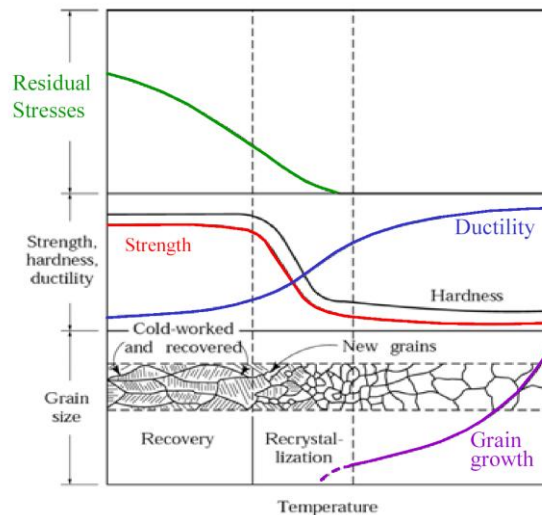
17

17



POLITECNICO
DI TORINO

Changes of Materials properties during hot rolling



04/12/2025

Mechanical Engineering

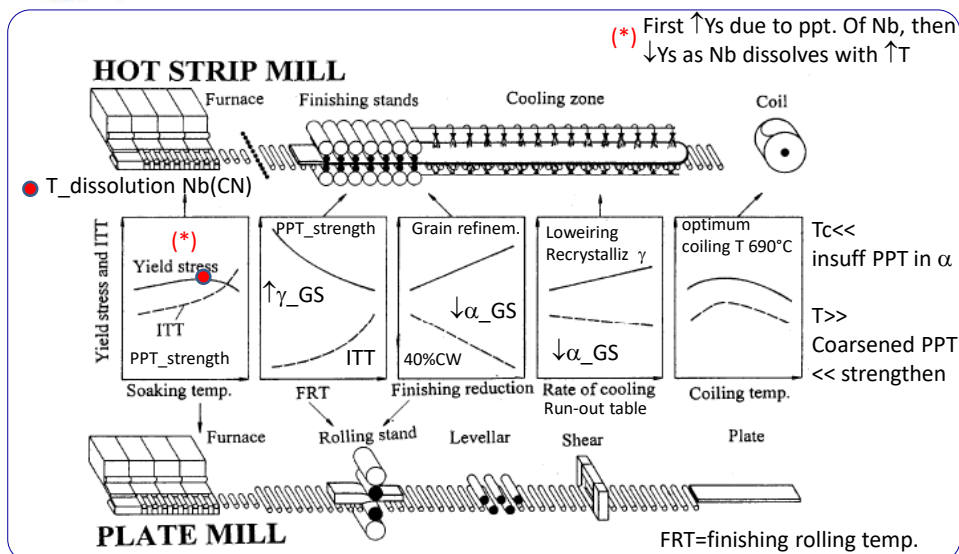
18

18



POLITECNICO
DI TORINO

TMCP of HSLA +Nb(CN)



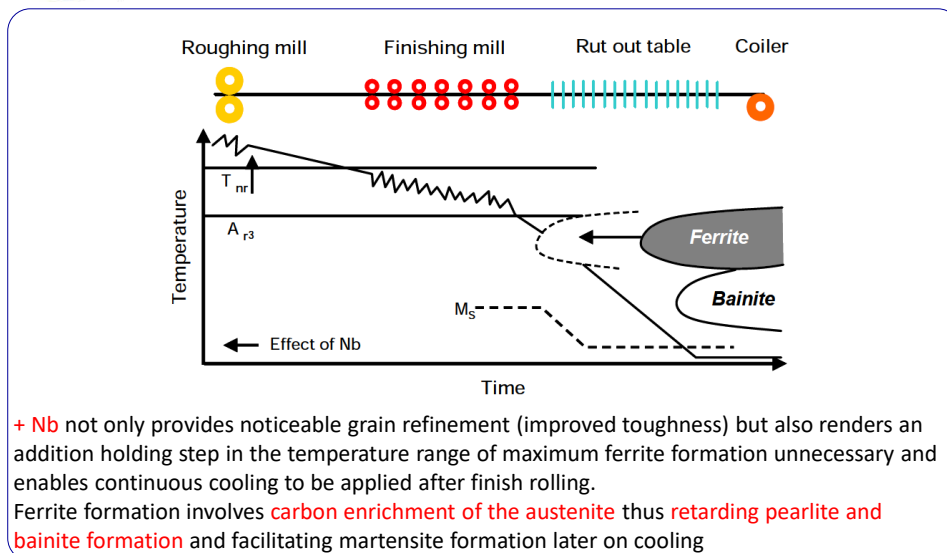
Effect of hot rolling parameters on mechanical properties during/after hot rolling (strips, plates)

19



POLITECNICO
DI TORINO

Hot Strip Processing of DP Steels: effect of Nb



Bleck ebook

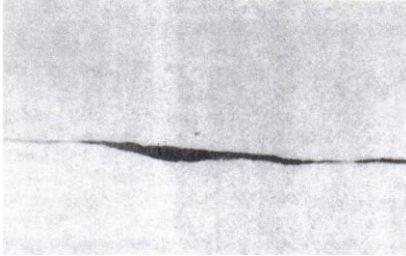
20



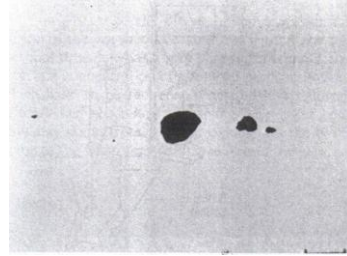
POLITECNICO
DI TORINO

Casting Defects: Inclusions

Fig 7-3



Non metallic elongated inclusion of MnS in steels



Non-metallic globular inclusion formed upon adding a special *spheroidizing* element in the melt

04/12/2025

Metallic Materials and Technology - Alloy
Shaping from Liquid Phase

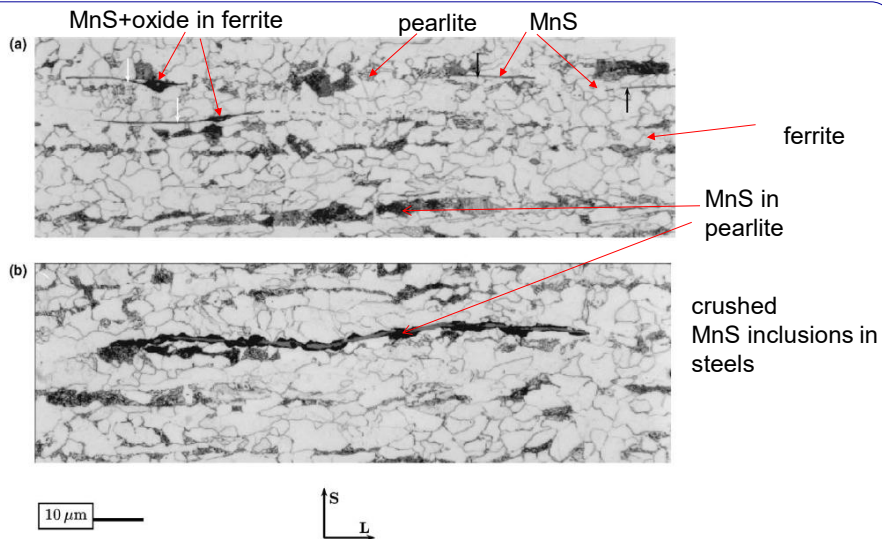
21

21



POLITECNICO
DI TORINO

Typical Hypo-eutectoid Steel Microstructure after rolling



04/12/2025

Mechanical Engineering

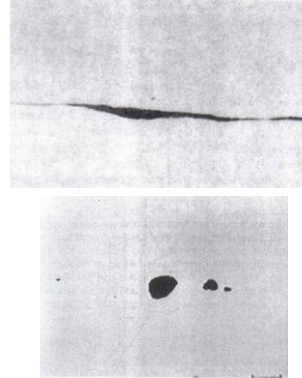
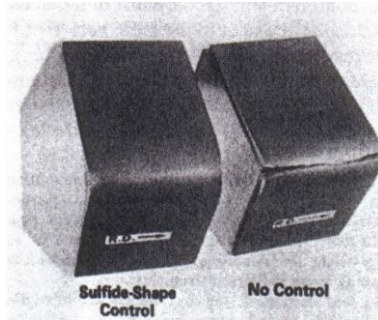
22

22



POLITECNICO
DI TORINO

Inclusion effects in Sheet Bending



Formability of sheets affected by spheroidal and elongated (high anisotropic) MnS inclusion.

04/12/2025

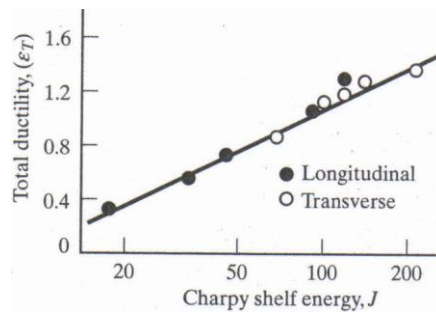
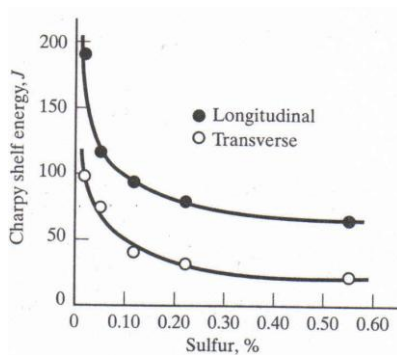
23

23



POLITECNICO
DI TORINO

Inclusions effects



Non metallic inclusion effects on the upper energy shelf of an impact energy test curve (CVN) as well as on overall ductility.

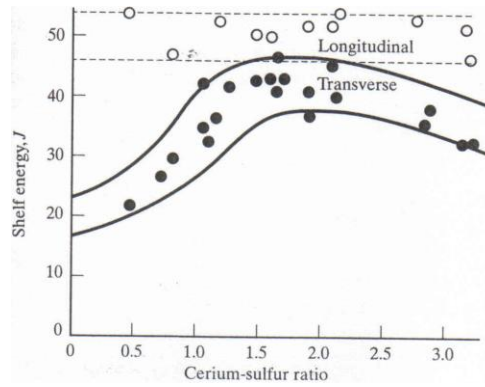
04/12/2025

24

24



Inclusions effects



Influence of rare earth elements (Cerium) on impact test of samples taken along a transversal direction due to globular inclusions.

25

04/12/2025